AI-TPMS PROJECT STATUS & DEVELOPMENT ROADMAP

System Health: 100% Operational | 15 Test Suites Passed | Production Ready

1. Current Module Completion Status

Subsystem / Module	Status	Test Coverage	Verification Notes
Authentication & RBAC	COMPLETE	100% (4 Tests)	JWT tokens, role guards (Student, Trainer, Admin).
Resume ATS Engine	COMPLETE	100% (4 Tests)	PDF extraction, symbol parser (C++, C#, SQL).
3-Round Sequential Gating	COMPLETE	100% (5 Tests)	Strict score gating (Round 1 ≥ 65%, Round 2 ≥ 70%).
Computer Vision Proctor	COMPLETE	100% (3 Tests)	YOLOv8 cell phone detection & multi-person alerts.
MediaPipe 3D Head Pose	COMPLETE	100% (3 Tests)	Euler angle estimation (Pitch, Yaw, Roll, Gaze).
Speech & Audio Analyzer	COMPLETE	100% (3 Tests)	WPM calculation, hesitation pauses, filler words.
Mitra AI Mentor	COMPLETE	100% (3 Tests)	Dual-engine (Gemini Live + 115KB Offline KB).
Executive Analytics & PDF	COMPLETE	100% (3 Tests)	Radar charts, ReportLab PDF generation, weak-spot advice.

2. Strategic Engineering Roadmap

- **Phase 1 (Achieved):** Core API engine, 47 corporate job drives, 3-round sequential progression, and UI integration.
- **Phase 2 (Achieved):** Mitra AI Mentor widget with responsive carousel, offline knowledge base, and live voice input.
- **Phase 3 (Next Steps):** Multi-college multi-tenancy, enterprise LDAP synchronization, and WhatsApp placement notification webhooks.